

JAIKRISHNA (JAI) V. RAGHAVAN

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EDUCATION

University of Maryland

B.S. Computer Science - Cybersecurity Track, Minor in Linguistics

GPA: 3.73

Honors College, University Honors Citation

College Park, MD

May 2025

November 2023

TECHNICAL EXPERIENCE

Senseonics

Software Engineering Intern

Germantown, MD

Jun 2024 - Aug 2024

- Tested and resolved errors in **multi-threaded .NET applications** to reduce failure rate by 20%.
- Developed an XML-configurable User Interface for automating quality control procedures at production sites.
- Worked in an agile development environment for stakeholders in process engineering and software quality
- Optimized validation procedures in playlist loading scripts to reduce validation errors by 25%.

Senseonics

Software Engineering Intern

Germantown, MD

Jun 2023 - Aug 2023

- Developed the frontend of a Quality Control automation app using **C#, Visual Studio, and .NET**
- Created several interactive forms to process user input and execute commands from a .CSV file.
- Applied principles of **object-oriented design** to encapsulate UI components.

Senseonics

Software Engineering Intern

Germantown, MD

Jun 2022 - Jul 2022

- Refactored an **ASP.NET** web application to generate a .CSV file with live data for over 10,000 transmitters.
- Migrated transmitter firmware version data from a **Microsoft SQL Server** to the application.
- Enabled categorization of data by product type to improve process development data analytics.

SKILLS AND COURSEWORK

Programming:	JavaScript, Python, C#, C/C++, Ruby, OCaml, Haskell, Java, Kotlin
Tools/Frameworks:	.NET, Node.js, Express, Django, React.js, MongoDB, Git, Linux, Docker, Jira
Coursework:	Compilers, Computer & Network Security, Computer Systems Architecture Mobile App Development, Advanced Data Structures

PROJECTS

Nonogram Game Player and Solver - Django, React.js, SQLite

- Created a playable Nonogram game and integrated a Nonogram solving algorithm for generating and solving grids.
- Used **Django** for the backend and **React.js** for the frontend.

Fantasy Football Data Lookup - MongoDB, Express, React.js, Node.js

- Developed a MERN fullstack web application that allows users to search for fantasy football data.
- Accessed NFL player game and season usage data using ESPN's **REST API** endpoint.
- Leveraged **Node.js** and **Express** for the backend API and **MongoDB** to store player data.

MiniDafny - Haskell

- Implemented a subset of Dafny, a verification-based programming language, in **Haskell**.
- Processed input text with an **applicative parser** to generate an abstract syntax tree.
- Generated **verification conditions** for each statement by calculating the weakest preconditions according to Hoare Logic rules. Passed verification conditions to Z3, a Microsoft SMT solver, to verify programs.

HONORS AND DISTINCTIONS

National Merit Scholarship

National Merit Scholarship Corporation
Sep 2021 - May 2025

Dean's List

University of Maryland
Spring 2022 - Present

Alpha Lambda Delta Honor Society

University of Maryland
Oct 2022 - Present